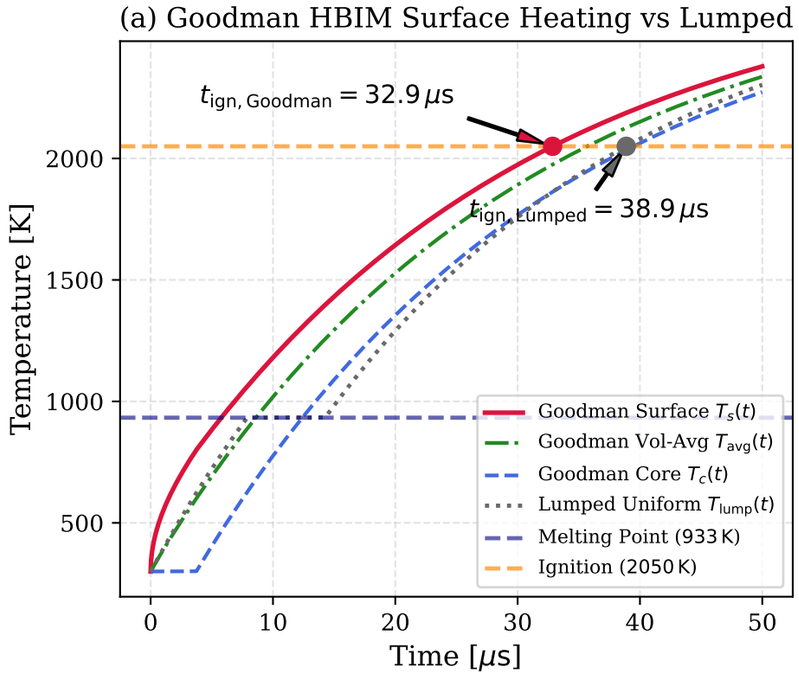
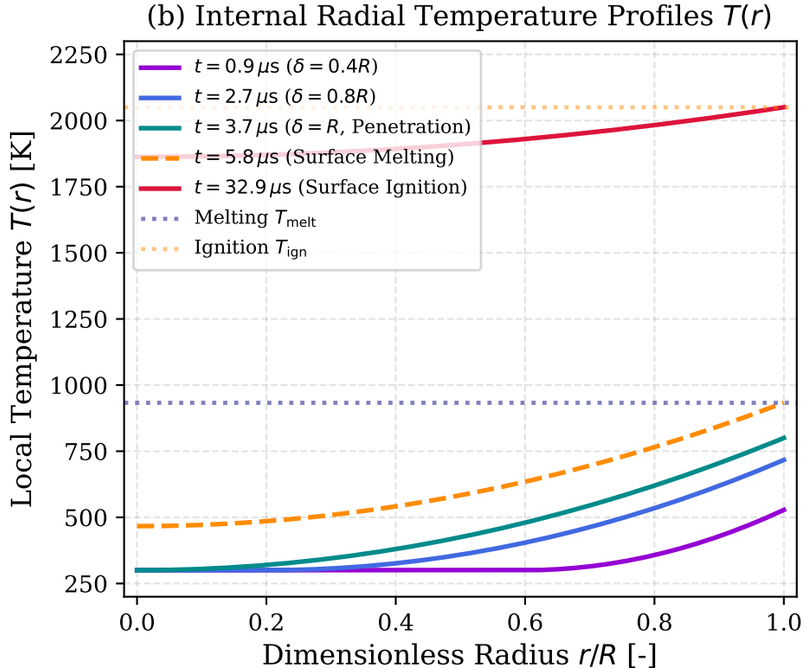
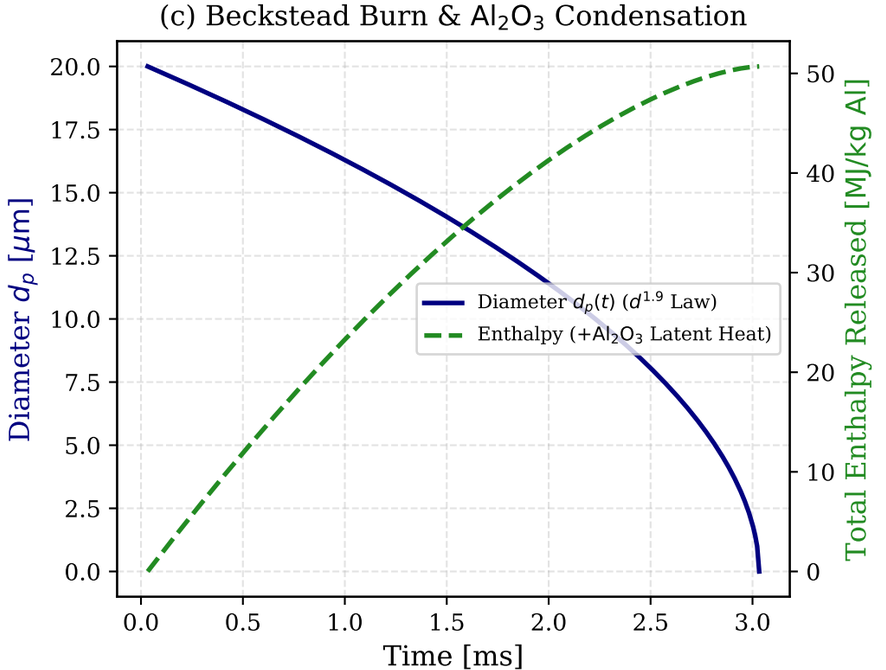

(a) Goodman HBIM Surface Heating vs Lumped
$t_{ign,\,Goodman} = 32.9\,\mu s$
$t_{ign,\,Lumped} = 38.9\,\mu s$
Goodman Surface $T_s(t)$
Goodman Vol-Avg $T_{avg}(t)$
Goodman Core $T_c(t)$
Lumped Uniform $T_{lump}(t)$
Melting Point (933 K)
Ignition (2050 K)
Temperature [K]
Time [μs]


(b) Internal Radial Temperature Profiles $T(r)$
$t = 0.9\,\mu s$ ($\delta = 0.4R$)
$t = 2.7\,\mu s$ ($\delta = 0.8R$)
$t = 3.7\,\mu s$ ($\delta = R$, Penetration)
$t = 5.8\,\mu s$ (Surface Melting)
$t = 32.9\,\mu s$ (Surface Ignition)
Melting T_{melt}
Ignition T_{ign}
Local Temperature $T(r)$ [K]
Dimensionless Radius r/R [-]


(c) Beckstead Burn & Al_2O_3 Condensation
Diameter $d_p(t)$ ($d^{1.9}$ Law)
Enthalpy (+Al_2O_3 Latent Heat)
Diameter d_p [μm]
Total Enthalpy Released [MJ/kg Al]
Time [ms]